

Notes on National Income Accounting and Balance of Payments

Author of this Note: Jayan Jose Thomas, 10 September 2024
Please do not circulate without the author's permission

The Circular Flow Of Incomes

Production of goods (say, food grains, steel or garments) and services (say, hair cutting or architectural services) **generates value**. One kilogram of rice will be of immense value for a family living in a food-insecure region — it could mean three or four hunger-free days for the children.

Production of goods and services **generates incomes** in the form of wages for workers or profits for the owners of firms.

One of the ways in which the value of a good or service is measured is by the price it commands in the market. The value of a haircut is often seen as the price paid to the hair stylist for the hair-cutting service.

In a market economy, the value is realized when the goods and services are exchanged (sale and purchase) in the **market**. The purchase of goods and services leads to **expenditures** in the economy.

We can see that production leads to the **supply** of goods and services. Expenditure generates **demand** for the goods and services supplied.

To complete a cycle of production-sale-purchase, both production-supply and expenditure-demand are critical.

If the T-shirts produced this month are not sold due to lack of demand, the garment manufacturer-seller may decide to reduce, or even stop, his sales and production activities next month. When the garment factory had to be closed last year after the workers left for their villages to celebrate Holi, there was a supply shortage (of T-shirts) in the market. An economy's growth can slow down due to **demand or supply constraints**.

The Economy and Social Classes

In a typical society, there are two classes of persons: workers who provide **labour** and owners of **capital**.

Workers do not own capital or other means of production. A landless agricultural worker cultivates the fields the landlord owns as he does not possess land or even agricultural tools. Not very different is the case of an employee in a multinational company who, despite her extraordinary skills and education, does not possess the networks or the financial resources to start her own firm.

Workers receive wages or salaries in return for the labour they provide. The owners of capital receive rewards in various forms depending on the nature of the capital they own: profits (for business enterprises), rent (for owners of land or buildings), interest (for creditors or holders of bank deposits), and dividends (for those who own equity). Unless otherwise specified, we shall refer to all these forms of payments to capital as 'profits'.

One of the most significant conflicts in any society is between labour and capital, between the interests of the working classes and those of the capitalists. Conflicts usually arise with questions relating to the division of income between

classes. Of the total income, what part should go to labour as wages, and what part should remain with the capitalists as profits? Such class conflicts often shape social and political relations in a society.

The winners and losers of labour-capital conflicts are not determined by economic factors alone. Wages and profits are not always proportionate to the economic contributions of labour and capital, respectively. Historical, social and political factors also come into play. In rural India, women workers receive lower wages than male workers, mainly as an extension of the relative disadvantages women face in society.

How does the Circular Flow Operate?

To begin with, think of the economy as a small island that does not engage in trade or other relations with the rest of the world. It is a closed economy. Assume that there is no government on this island. The economy comprises capitalists (or owners of capital or firms) and workers. The workers supply labour to the firms, and the firms organise the production of different types of goods and services. The workers receive wages in return for their labour, and the capitalists make profits (which can be seen as a reward for their entrepreneurship).

The value of **production** is realized when the households (of workers and capitalists) **purchase** the goods and services using the **incomes** they received for their role in the production process.

Production of goods and services generates **incomes**. Incomes are required for **expenditures** (purchase of goods and services). Expenditures are necessary for the firms to continue production (of more goods and services).

One of the first lessons in national income accounting:

The value of all goods and services produced in a country in a particular period
= Incomes generated in the course of this production **= Expenditures**
incurred in the purchase of these goods and services

Here, we can see a circular flow from production to income **to expenditures**.
See the Figure below.

GDP is Quantities times Prices

What is **GDP** or Gross Domestic Product? GDP is the sum of the value added by the production of all goods and services in a country during a given period (typically a year or a quarter).

GDP is the sum of incomes received by persons who participate in the production process (as workers or as capitalists). GDP is also equivalent to expenditures on the goods and services produced in the economy.

Value added refers to the value of output minus the value of inputs.

How do we measure value? As already noted, in a market economy, the value of a good or service is measured by the price it commands in the market. Therefore, GDP is estimated from two distinct and equally important economic variables. One is **quantity**: quantities of goods and services that are produced. The other is the **price**: prices of these goods and services in the market.

If X_1 is the quantity of good/service 1 and P_1 is its price, and so on...

$$GDP = P_1 X_1 + P_2 X_2 + \dots + P_n X_n$$

The above estimation method — quantities times prices — allows us to aggregate (or add up) goods and services that are very different in their nature: tonnes of steel, litres of milk, numbers of haircuts, hours of teaching, and so on.

What enters the GDP Calculations, and What does not?

A number of goods and services that are essential for society and the economy do not enter the GDP calculations as they are not traded in the market. **Childcare or cooking** are two good examples. If a **family member** offers these services, they will NOT be counted as part of GDP. However, if paid helpers offer the same services, they WILL BE counted as part of the GDP estimates.

Nevertheless, the practice is different in the case of agriculture and allied activities. A farmer's production of **food grains for consumption** within her own household **gets reported** in the national income estimates.

Approximately half of all adult women in India are neither employed nor unemployed (and seeking jobs), but instead, they attend to domestic duties within their own homes (child care, cooking) (according to official statistics; see below). Without a doubt, most of these women work long hours within their own households. However, these women are considered to be outside the labour force, and their services are not counted in the GDP numbers. It is also true that the society undervalues women's contribution.

In East Asian countries, large numbers of women have found jobs in the manufacturing and services sectors. Along with these changes, food courts and

child-care service centres have sprung up across these countries. Therefore, the services performed within the households, typically by women, have begun to be offered by the market. The GDP figures in these countries receive a double boost: first, from women's work in the factories and offices (and not in their homes); second, when cooking and other services performed within the homes began to be supplied by the market.

.....

In India, official statistics suggest that **women's participation in the labour force** is very low. In 2022-23, among women aged 15 years and above in India, only 36% were employed, while 46% reported that they were engaged in domestic duties (such as cooking and child care) within their own homes. In comparison, the corresponding proportions of the employed and those engaged in domestic duties among men (aged 15 years and above) were 76% and 0.8% respectively. (India's Periodic Labour Force Survey (PLFS), 2022-23)

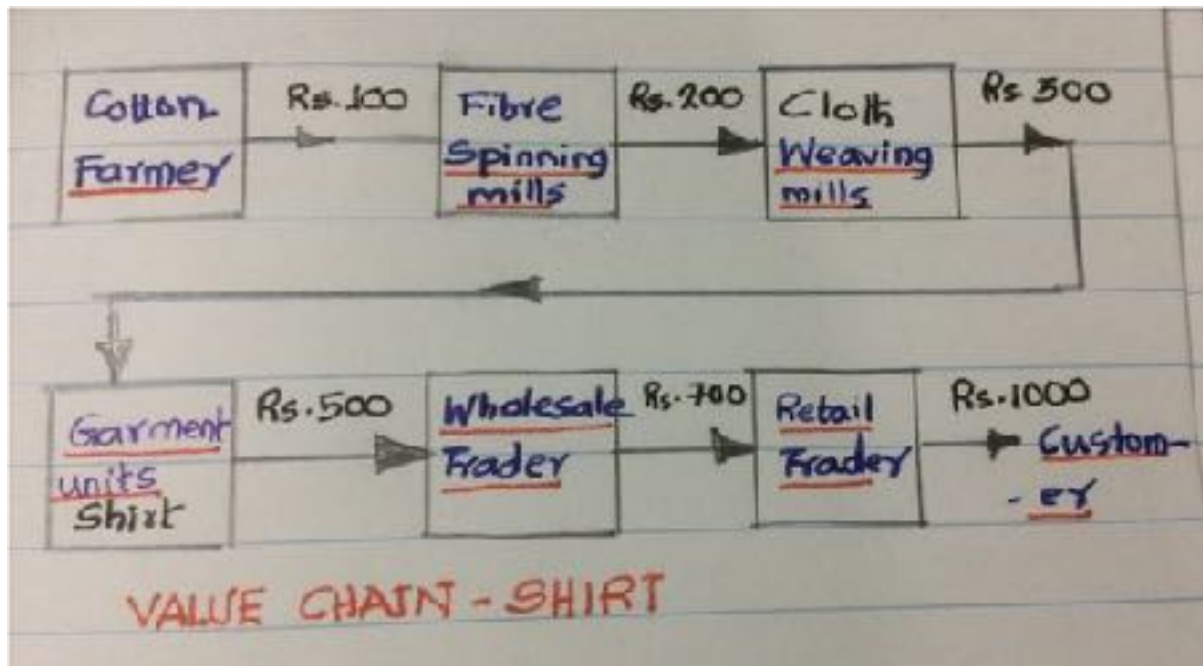
.....

Illegal activities or under-reported incomes do not enter the national income estimates.

At the same time, it is to be noted that **informal-sector economic activities** (such as street-side shops or small manufacturing units) are very much included in the national income estimates.

Value Added is Output minus Input
GDP as Sum of Value Added, GDP as Final Expenditures

Consider the chain of economic activities that lead to producing and selling a piece of garment (a T-shirt). The first stage in the production process is the production of cotton by farmers. The cotton is then processed, first in spinning mills, then in weaving mills, and finally in garment-making units. The wholesale



trader purchases the shirt from the factories and sells it to a retail shop. The retail shop sells the shirt to the customer for Rs.1000.

There are two ways to calculate the GDP:

GDP as the **sum of value added** by individual sectors =

Farmer (Rs. 100) + Spinning mills (Rs. 100) + Weaving mills (Rs. 100) +
Garment units (Rs. 200) + Whole sale traders (Rs. 200) + Retail traders (Rs.
300) =

Rs. 1000

(The value added by each sector = Output — Input)

Or GDP as the value of **final** output = Rs. 1000

Goods and services produced are of three types:

Intermediate goods and services are used as inputs for producing other goods and services (during the same period in which they were produced) (eg. chemicals).

Consumer goods and services that are consumed directly (eg. food products, garments, shoes)

Capital goods refer to machinery and other fixed investment goods needed in the production process (once installed, machinery can be used in the production for several future years) – these are ‘machines that produces machines.’

We assume that the wages received by workers are spent entirely on consumer goods (we infer that workers do not save anything!). Intermediate goods are shared among firms. Owners of firms invest their profits entirely in capital goods (which will help them improve their production and profit in future years) (we also assume that the owners of firms do not consume anything, or more realistically, their consumption is a negligibly small share of their incomes!).

The Circular Flow of Incomes

Production of Output	Incomes generated in the course of production	Expenditures incurred in the purchase of goods and services
Consumer goods and services	Workers receive Wages	Consumption (of consumer goods and services by workers)
Capital goods and services	Owners of capital receive Profits	Investment (in capital goods by owners of firms)
Intermediate goods and services (or inputs)	Payments for inputs (firms pay each other)	Payments for inputs (firms pay each other)

We can see that this is a circular flow: the act of production generates incomes; these incomes are spent to purchase the goods produced.

Value of Output = Incomes = Expenditures

Output = Input + Wages + Profits (or Incomes for owners of capital)
Output = Input + Consumption + Investment

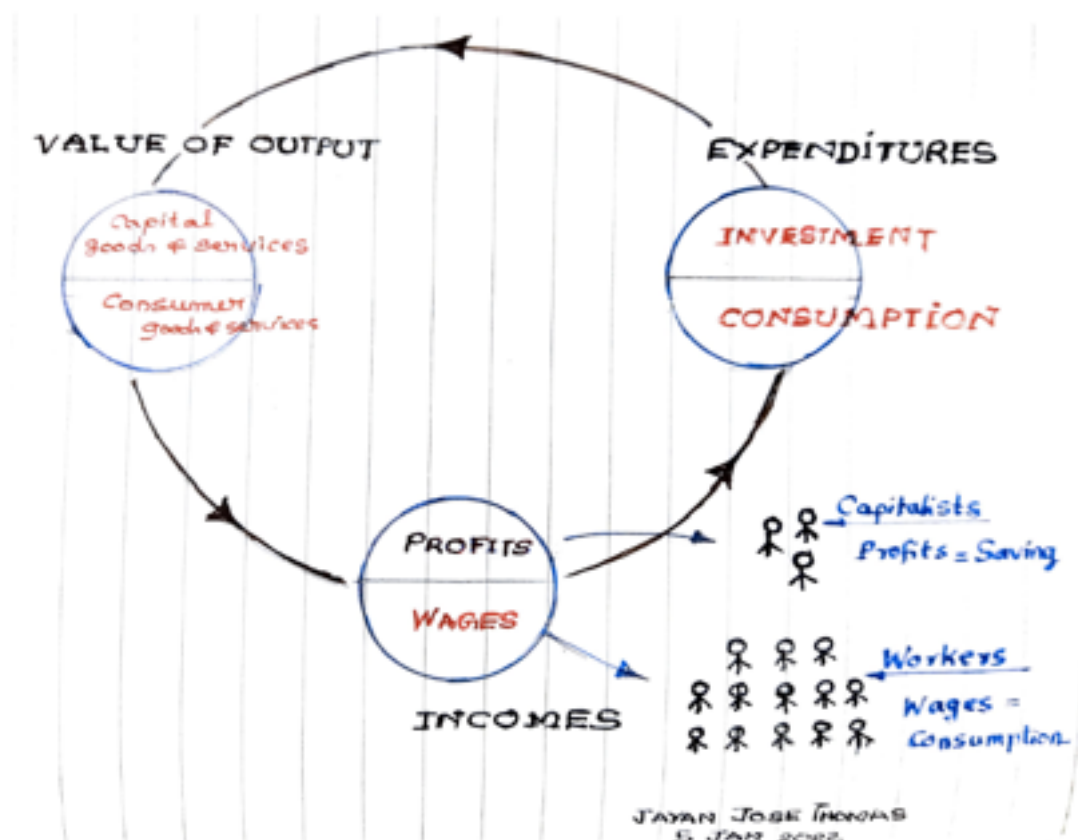
GDP is the sum of value added in the production of all goods and services.

GDP = Output – Input

Therefore in a closed economy with no government

GDP (as expenditures) = Consumption + Investment or
= C + I;

Also GDP (as incomes) = Wages + Profits;



GDP at Current (or nominal GDP) and GDP at Constant Prices (or real GDP)

As we know, output and GDP are expressed in value terms (such as in Rupees or dollars). Consider that X_i is *output net of input* of the i^{th} good or service and P_i is its price. Therefore,

$$\text{GDP} = \sum_i P_i X_i$$

(Note: In the earlier Section we used X to denote output. In this section, however, X is the symbol for output minus (or net of) input. The change in symbol is only for ease of presentation).

The above can lead to a problem when we compare, for instance, GDP in 2019, which is production in 2019 times prices in 2019, with GDP in 2020, which is production in 2020 times prices in 2020. How much of the increase in GDP between 2019 and 2020 is due to the actual increase in production (increase in the production of *real* goods and services), and how much of it is just a result of the rise in the prices of goods and services between the two time periods?

This problem can be resolved if we obtain the GDP for both years at constant prices or at the prices of the base year (say, 2011). The statistical agencies choose a base year from time to time, depending on a number of considerations.

GDP at constant (2011) prices for 2019: this will be production in 2019 times prices in 2011.

Increase in GDP between two years = GDP_{2019} to GDP_{2020} :

How much of this increase is due to an increase in prices, and how much of the increase is due to increase in real quantities

Nominal GDP for 2019:

$$P_1(2019) * X_1(2019) + P_2(2019) * X_2(2019) + \dots + P_n(2019) * X_n(2019)$$

Real GDP for 2019 or GDP at constant prices (with prices in 2011 as the base year):

$$P_1(2011) * X_1(2019) + P_2(2011) * X_2(2019) + \dots + P_n(2011) * X_n(2019)$$

Measuring Price Changes: GDP Deflator, WPI and CPI

(assuming 2011 as the base year)

GDP Deflator = Nominal GDP/Real GDP

$$= \sum_i P_i(2019) * X_i(2019) / \sum_i P_i(2011) * X_i(2019)$$

(Paaschi's Index)

WPI (Wholesale Price Index) and CPI (Consumer Price Index) are specifically designed to understand the growth of prices. The government statistical agencies track the changes in prices for a *specified basket of goods and services* over the years.

$$\text{WPI and CPI} = \sum_i P_i(2019) * X_i(2011) / \sum_i P_i(2011) * X_i(2011)$$

(Laspeyres Index)

The above shows the price index for 2019 with 2011 as the base year: how much have the prices of a basket of goods (the weights for the basket assigned in 2011) changed between 2011 and 2019?

WPI tracks price changes affecting producers, especially of manufactured products. CPI tracks price changes affecting consumers in rural and urban areas. In India, there are separate price indices to understand price changes affecting the households of agricultural labourers, industrial workers, urban non-manual employees, and so on.

.....

The WPI attaches relatively high weights to various manufactured products, such as basic metals and chemicals. Food and beverages have a weight of 54.2% in CPI-rural but only 15.3% in WPI. Housing (rental expenses, and so on) has a weight of 21.7% in the CPI for urban areas, but no weight is attached to this group in rural areas.

Weights for WPI in India, Base year: 2011-12 = 100

Sl. No	Major groups	Weight
1	Food articles	15.26
1.1	Cereals	2.82
2	Non-Food primary articles	4.12
2.1	Crude petroleum	1.95
3	Fuel and power	13.15
3.2	Petrol	1.6
4	Manufactured products	64.23
4.2	Chemicals	6.47
4.3	Basic metals	9.65
	All groups	100

Weights for CPI in India, Base year: 2012 = 100

Sl. No	Major groups	Weights	
		Rural	Urban
1	Food and beverages	54.18	36.29
1.1	Cereals and products	12.35	6.59
1.2	Meat and fish	4.38	2.73
1.3	Milk and products	7.72	5.33
2	Pan, tobacco and intoxicants	3.26	1.36
3	Clothing & footwear	7.36	5.57
4	Housing	0	21.67

5	Fuel and light	7.94	5.58
6	Miscellaneous	27.26	29.53
6.2	Health	6.83	4.81
6.3	Transport and communication	7.6	9.73
6.5	Education	3.46	5.62
	All groups	100	100

Source: <https://pib.gov.in/PressReleasePage.aspx?PRID=1623753>

.....

Income, Capital (Or Wealth), And Investment

A and B are two workers in a firm that pays them a monthly salary of Rs. 25,000 each or an annual salary of Rs.300,000. Therefore A and B receive the same incomes.

While being equal in their income levels, A and B can be quite unequal with respect to their ownership of wealth. A owns a flat valued at over Rs.6 million, an asset passed on to him as an inheritance from his parents. B comes from an impoverished family, and her salary is just sufficient to meet her monthly expenditures. She lives in a rented flat, and the value of all her belongings does not add up to even Rs. 100,000.

Capital or wealth that people hold includes factories, infrastructure, machinery, patents, and so on (these are productive or professional capital — because they directly aid the process of production). Wealth also includes all forms of real property (including residential real estate) as well as financial assets. Capital stock (mainly machinery and factory buildings) may be seen as a subset of wealth.

How is income different from capital (or wealth)?

We know that:

Income = Value of output (commonly referred to as GDP) (produced within the country) = Expenditure

(National) Income = Labor income + capital income

Labour income is in the form of wages or salaries. Capital income can be in the form of profit (from operating a factory or business) but also as rent, dividend, and interest (the last two are returns on financial assets).

Expenditure = Consumption + Investment (assuming, for simplicity, no government and no trade with the external world).

Investment is the addition to capital stock.

Consider the case of a factory. The capital stock owned by the factory in a particular year (say 2023) is the value of all machines and the factory building in 2023. We are referring to the cumulative stock of machines purchased and buildings built in all previous years and in 2023.

In 2024, new machines have been purchased, and a few new wings have been added to the factory building. These form the investment for 2024. Investment in a particular year (2024) in machines or factories has added to the existing capital stock (of machines, factory buildings, or houses).

Income and investment are flow variables: Continuous flow of incomes or investment (like a stream or a river).

Capital is a stock variable (like a large lake or the sea — to which the stream or the river joins).

But we also have to consider the depreciation charges — the cost of maintaining the buildings and keeping the machines free of wear and tear.

We denote K as capital stock, I as investment, δ as depreciation

$$K_{2024} = K_{2023} + I_{2024} - \delta_{2024}$$

Capital stock in 2024 is equal to the capital stock in 2023 plus investment in 2024 minus the depreciation charges incurred in 2024.

The national income statistics estimate the depreciation charges incurred by the economy as a whole.

**Net Domestic Product (NDP) =
Gross Domestic Product (GDP) — Depreciation**

.....

In the example at the beginning of this section: while the capital or wealth owned by A is 20 times his annual income, the ratio of capital to income for B is less than 1/3. If we denote capital as K and income as Y , then the K/Y ratio is 20 for A and 1/3 for B.

The K/Y ratio can be estimated for the economy as a whole. Thomas Piketty, in his book *Capital in the Twenty-First Century*, writes that unequal societies are characterized by K/Y ratios of 6 or more.

.....

The Relation Between Saving And Investment:

GDP = Value of Production = Income = Expenditure

In a simple two-sector (firms and households) economy (with no government and no trade with the external world):

$$\begin{aligned}\text{GDP as Income} = Y &= \text{Wages} + \text{Profits} \\ &= \text{Consumption (C)} + \text{Savings (S)}\end{aligned}$$

A simplifying assumption is that wages are entirely consumed and profits are fully saved. In reality, for most of us, the incomes we receive are partly consumed (C) and partly saved (S). In a country like India, the wages received by a majority of workers are too low, and much of it is used for daily expenses, leaving only a little for saving. So most of the wages are used for consumption. At the same time, for earners of profit, daily consumption expenditures may be a small share of their earnings. So a large part of the profits are saved.

$$\text{GDP as Expenditure} = Y = \text{Consumption (C)} + \text{Investment (I)}$$

The above implies that Savings (S) = Investment (I) (Savings Investment identity)

.....
Now let us look again at the Income = Expenditure and Saving = Investment identities (S = I). As savers and investors are not the same set of individuals, the economy will likely face a crisis in the long term. Will all the saving in the economy translates itself into investments? What if some of the capital owners prefer to keep their savings in the form of gold or some other assets that do not reach potential entrepreneurs and perhaps never fructify into investments?

Who save? Who invests? Who are the entrepreneurs

Some say that they have ‘invested’ money in the stock market, the bond market, and so on. But in reality, they are not ‘investing’ but saving – keeping their savings in stocks or bonds.

From a macroeconomic perspective, investments refer to setting up a factory or starting a new business enterprise (‘brick and mortar’ investments) — typically activities that lead to the production of more goods and services in the economy.

Entrepreneurs who undertake such investments are willing to take the associated risks.

The savings required for investments can come from profits (reinvested by firms), equity, bonds or bank deposits. There could be millions of savers in any economy, including big businesses, rentiers, holders of stocks, bonds or bank deposits (A part of the savings required for investment may flow from the incomes of workers who save in the form of bank deposits or equity shares).

However, the entrepreneurs who invest — who set up businesses or establish factories — are far fewer in number. Not all capital owners possess the skills, the vision and the perseverance required to be successful as an entrepreneur.

Banks and other financial institutions (such as stock markets and bond markets) help to mobilize the savings of individuals whose expenditures are less than their incomes. These financial institutions also help to channel such savings to potential entrepreneurs — individuals or firms who use these savings (made available to them in the form of bank credit, public holding of equities issued by the firm, and so on) to make investments. (Nowadays, in many sectors, one does not have to have a lot of savings to start an enterprise. Start-up funds or bank loans will be available to potential entrepreneurs with good business ideas).

How do Demand Shortages Lead to an Economic Slowdown?

The numbers in the paragraphs below are imaginary. They are meant to provide an illustration of the concepts; there is no need for you to spend much time on the numbers per se.

Assume that the **planned** expenditure for the year 2020 was Rs.300 tr (trillion). Note the word ‘planned’. In 2019 (or even earlier), the owners of factories and

businesses anticipated that the aggregate expenditure or demand in the economy would rise to Rs.300 tr in 2020. Given such expectations about future **demand**, they purchased (in 2019 itself) new machinery (wherever needed), hired workers, and set up new showrooms to increase production (and, therefore, supply).

So, the planned expenditure for 2020:

300 tr = 180 tr as consumption expenditure + 120 tr as investment expenditure.

During the first three months of 2020, the expenditures (demand) were behaving in line with expectations. Enthused by the demand, the factories produced output vigorously, generating sufficient stocks to last for the coming months. But then the covid-19 pandemic hit all of a sudden. After the government announced the lockdown in March 2020, the factories had to stop production, at least for a few weeks; many workers lost jobs and incomes, and there was uncertainty and gloom everywhere. On account of the pandemic and the lockdown, there were supply constraints. Still, more seriously, there were demand shortages following the loss of incomes for vast sections of India's informal workforce. There were not enough buyers for the buildings, the machines and the goods that were already produced

Given the fall in demand and the overall uncertainty, many entrepreneurs lost the confidence to continue investing in the economy. By the end of the year, the actual investment expenditures in 2020 amounted to only Rs.100 tr, or Rs.20 tr less than what was originally planned. The unsold machines and construction materials ended up as unplanned additions to inventory in the godowns.

Thus, supply or production in 2020 amounted to Rs.300 tr, but the actual demand was only for Rs.280 tr worth of goods and services. However, for accounting purposes, **addition to inventory is also considered investment** (unplanned inventory investment).

So, the demand structure for 2020 was the following:

Rs.180 tr of consumption expenditure + Rs.100 tr of investment+Rs.20 tr of inventory (for a total production of Rs.300 tr)

However, the fall in demand and inventory accumulation in 2020 would further slow the economy in 2021. The planned investment expenditure for 2021 was only Rs.90 tr. The entrepreneurs lowered their expectations about future expenditures by 20 tr because of the unsold inventories and by another 10 tr because of the continued demand uncertainties. But that was not all. The reduction in investment expenditures led to further loss of jobs for workers and a further fall in incomes and consumption expenditure in the economy.

The planned expenditure for 2021 was Rs.260 tr = Rs.170 tr of consumption expenditure + Rs.90 tr of investment expenditure.

Therefore, a fall in investment expenditure can lead to a downward spiral of expenditures and demand that can slow economic growth for several months or even years:

Fall in investment — inventory accumulation — further reduction of investment — loss of jobs and incomes — reduction of consumption expenditures — reduction of savings too — a slowdown of economic growth.

In other words:

$$\text{GDP as expenditure}_{2020} = C_{2020} + I_{2020}$$

$$\text{GDP as income}_{2020} = C_{2020} + S_{2020}$$

$$\text{GDP as expenditure}_{2020} = \text{GDP as income}_{2020}$$

C, I and S are consumption, investment, and savings respectively.

When I_{2020} was lower than what had been anticipated (at the beginning of 2020), the expenditure or demand in the economy fell to a level lower than usual. As a result, incomes and, consequently, saving fell to levels lower than usual. The crisis negatively impacts investments and expenditures in the coming years as well.

So the direction of macroeconomic changes is from **investment to saving in an economy that faces a demand problem.**

.....

On the other hand, think of an economy in which there is no demand problem. Factories are running at full capacity, and all workers are employed. In such an economy, if the saving supply can be increased – more productive capacity can be created (more factories can be built, more workers can be employed). That leads to more investment, more spending (or demand) and more incomes in the economy.

So the direction of macroeconomic changes can be from **saving to investment in an economy that faces a supply problem.**

Therefore, an important question in macroeconomics is: **does saving lead to investments, or is the direction of causation from investment to saving?**

In an economy suffering from demand shortages, the injection of demand by an external agency — the government — can restore demand and revive economic growth to a great extent.

Income And Distribution:

Will Wage Increases Help A Demand-Constrained Economy?

We have assumed that workers consume all their incomes (wages) while capitalists save all the incomes they receive (profits). Such an assumption is not too far away from real-life experiences. In India and other countries, the poorer segments of the population have a much higher **propensity to consume** — they consume a relatively high share of their incomes.

We can see that in an economy that faces a demand shortage, the redistribution of incomes in favour of workers (through higher wages) is beneficial for economic growth.

Higher wage incomes -- lead to higher demand – therefore, higher consumption and investment expenditures – lead to higher incomes.

On the other hand, in an economy that faces supply constraints (and no demand shortage), a higher level of profits may actually benefit economic growth.

Higher profits -- lead to increased saving supply – more productive capacity -- higher incomes and expenditures.

The above relation can be expressed algebraically.

The output produced in the economy is X . This is the “real” output – say in tonnes of steel or numbers of shirt prices.

Output can also be expressed in value terms: production of goods and services in physical units (X) times their prices (P).

Steel production in millions of Rupees)

P is prices (or price deflator).

Output = Input + Wage Payments + Profits (payments to capital owners)

$$PX = aPX + wbX + \pi PX$$

X = “real” output; Y = real income; P = Prices; PX will be nominal income (or output);

Input-output coefficient (or Input/ X) = a ;
labour-output coefficient (No. of workers/ X) = b

Employment or number of workers = $L = bX$.

w = “The” (average) money-wage rate (or wages received per worker),
Wages = wbX

π is profit share or the share of income (PX) paid out as profits

aPX = Outlays (amount spent) for intermediate inputs; wbX = Wages; πPX = Profits

GDP = Output — Input = $(PX - aPX) = wbX + \pi PX$ or Wages (Y_w) + Profit (Y_π)

We assume that wages are used for consumption and profits (which equals savings) are used for investment.

C and I are consumption and investment in real terms (tonnes of food, steel,..);
 PC and PI are consumption and investment in nominal terms, that is, multiplied by their respective prices.

$$GDP = wbX + \pi PX = Y_w + Y_\pi = PC + PI$$

K is capital stock, I is investment, δ is depreciation (Remember: $K_{2021} = K_{2020} + I_{2021} - \delta_{2021}$)

PK is capital stock in value terms (in Rupees or dollars)

Return on capital or profit rate (**or r**) = Profit/Capital stock = $\pi PX/PK$

X/K or Output as a share of capital stock is an indicator of capacity utilization in the economy. We denote X/K as u

(You have a factory and machines worth Rs. 100 million. In 2019, this factory produced output worth Rs. 10 million. In 2020, however, output fell to Rs. 7.5 million. Capacity utilization may have fallen (say from 100% in 2019 to 75% in 2020).

The factory owner will be concerned about the profit rate – and not on the profit share of output.

We can see that profit rate, $r = \pi u$ (profit share times capacity utilization)

One can see that in a demand-constrained economy, an increase in wages will help economic growth.

Higher wages lead to more incomes for workers, greater demand for goods and services, and greater capacity utilization in the economy. u will increase. Therefore, even if π (profit share) remains stagnant or falls somewhat, profit rate (which is $r = \pi u$) will still increase.

.....

Real wage $\omega = \text{Nominal wage/Price} = w/P$ and the profit share = Π

According to the Keynes-Kalecki model:

The three variables — w , ω and Π — are influenced by historical and institutional factors and also by economic policies (of the government).

The levels of real wages (ω) and profit share (Π), together, provide an indication about the structure of distribution in the society-economy: what part of the overall income pie goes to the workers and what part to the capitalists? These variables too can be influenced by policies: consider social policies, including subsidized food and education, which may reduce the general cost of living and thereby increase real wages.

.....

Now The Government Enters Income Calculations

GDP is Value Added or Output minus Input.

In the Indian statistical system, this is referred to as:

Gross value added (GVA) at basic prices – the sum of the value of all goods and services produced in the economy – enough to make payments to the factors of production – and before paying taxes to or receiving subsidies from the government.

Gross value added (GVA) at basic prices

= Wages + Profits (or Capital Payments)
(both are factor incomes)

= Consumption expenditure + Investment expenditure

Now the government enters the picture. The government collects taxes for the goods that reach the market (from the factory). This is the government's

income. At the same time, a part of the taxes collected are redistributed as subsidies

GDP and GVA at basic prices

GDP = GVA at basic prices + Indirect Taxes – Subsidies (or transfers from government)

The government uses its tax incomes

(i) to be redistributed as subsidies to the needy (so we have food subsidies, education subsidies and so on) (or Transfers from the government)

(ii) for government consumption (payments to government employees....)

(iii) for government investment (in railways, dams, highways, and so on)

Now, we will have to consider government expenditure as well.

$X = C + I + G$

where C is private consumption, I is private investment and G is ($C_g + I_g$) or consumption and investment expenditures by the government

Subsidies or transfers from government do not enter GDP calculations. They are transfers from the government – not paid against any new good or service produced in the economy.

.....

More on:

Transfer payments (from government):

These are incomes in the form of old age pension or unemployment benefits that are obtained as a result of redistribution or transfer of primary incomes by the government.

Importantly, transfer payments are payments that are made not against any goods or services produced in the economy in the current period. In other words, these are payments for which nothing is paid in return.

As a result, transfer payments are excluded from national income estimates. It can be seen that including transfer payments in GDP estimations can result in double counting of incomes.

Consider that in 2014-15, a person P made an income of Rs.10000 (which was a payment she received for some good or services produced in the economy). Out of her income, P paid Rs.1000 as taxes to the government. Out of the Rs.1000 collected as taxes from P, the government paid an unemployment benefit of Rs.500 to Q. This was, therefore, a transfer of a part of P's income to Q. If we include the unemployment benefit of Rs.500 as part of national income, there will be a double counting of Rs.500 (It is already included as part of P's income).

.....

Income Calculations In An Open Economy

We will now consider the economy's interactions with the outside world. We export our goods (so a part of the domestically produced output is sold abroad) but we also import goods from abroad.

X is the domestic output *net of* domestic inputs.

$$\mathbf{X = C + I + G + E}$$

E is exports of (domestically produced) goods and services to the rest of the world

GDP is X minus M

M is imports of goods and services (produced elsewhere in the world)

GDP is the domestic output *net of* domestic inputs minus inputs from the rest of the world.

$$\mathbf{GDP = C + I + G + (E - M)}$$

and

What about incomes?

We know that GDP = Aggregate domestic incomes earned from the output produced within the national territory (regardless of whether or not it is produced by the country's residents). (Note: residents: not necessarily citizens)

The gross domestic product includes depreciation, which is an allowance made to cover the wear and tear of machines. **GDP minus depreciation is net domestic product.**

Gross National Income (GNI) = Aggregate incomes earned from the output produced by the country's residents (regardless of whether this production is occurring within the country's borders or outside it)

Therefore, **Gross National Income (GNI) = GDP + NFI**

Net Factor Incomes from abroad (NFI) = Income paid to domestic factors of production (labour, capital) by the rest of the world minus the income paid to the foreign factors of production by the domestic economy.

The multinational firm Hyundai produces cars in India. The income earned from Hyundai's automobile production in India will be part of India's GDP. If Hyundai repatriates a part of the profits from its India operations back to its home country, the amount of profits thus repatriated will be considered as an

outflow (with a negative sign) of factor incomes from India. That amount will be deducted while calculating India's GNP.

An Indian resident (who spends the bigger part of the year — more than six months — in India) works as a consultant on a short, three-month-long project with a US company. The income thus earned by this person will be counted in India's GNI, as an inflow (with a positive sign) of factor incomes from abroad.

At the same time, the **remittances sent to family members by a non-resident person** will not be counted in GDP or GNI estimations.

Consider a non-resident Indian who is a migrant worker in the UAE. The work carried out by this person is counted in UAE's GDP. As this person is not an Indian resident (she may be an Indian citizen and may have close family relations in India; but she is a long-term resident of UAE), her income cannot be included in India's GNI either.

.....

The Indian resident who is working on a short-term consultancy project in the US will have to report his foreign income while filing his income taxes in India. Of course, the taxes he already paid in the US will be deducted from his tax burden in India. At the same time, the non-resident Indian living in UAE does not have to report her foreign income to Indian authorities.

.....

At the same time, the remittances sent to India by a non-resident migrant worker is an inflow of foreign currency into India. Therefore, this remittance flow will appear in India's balance of payments.

The remittance income received by the family members is also part of the aggregate incomes received by Indians. It is in the nature of a transfer payment received from abroad (payment received not as a remuneration for the production of any good or service).

(Remember the class discussion on the differences between NFI and net transfers from abroad. Note that 'transfers from abroad' do not enter GDP or GNP, but NFI enters GNP).

Aggregate Income received by citizens of a country = GDP + NFI + Net Transfers from abroad (mainly remittances).

Saving- Investment Relation In An Open Economy With Government

Note in the above relation that $(E - M) + \text{NFI} + \text{Net Transfers from abroad}$ is Current Account Balance (or CA)

NFI is Net Factor Incomes from abroad

If $(E - M) + \text{NFI} + \text{Transfers from abroad}$ is +ve, the country's current account is in surplus. If it is -ve, the country's current account is in deficit.

Private disposable income (income at the disposal of private agents)
= Aggregate Income received by citizens - Taxes + Transfers from government ---(1)

But **Private disposable income is also equivalent to: Private Consumption (C) + Private Savings (S_{pvt}) ... (2)**

(1): Private disposable income (Y_P) = $C + I + G + (E - M) + NFI + \text{Net Transfers from abroad} - \text{Taxes} + \text{Transfers from government}$.

Where C is private consumption, I is private investment; G is Government Expenditure, E is export of goods and services; M is import of goods and services.

Equals

(2): Private disposable income (Y_P) = $C + S_{pvt}$

From (1) and (2), we can see that:

$$(I - S_{pvt}) + (G - T) + (E' - M') = 0$$

Or

$$I (\text{private investment}) = S_{pvt} + S_{govt} + S_{foreign}$$

.....
 $(E - M) + NFI + \text{Net Transfers from abroad}$ is $E' - M'$ or Current account balance or:

Inflows (of payments, say in dollars) to the country from the rest of the world (Let us denote it as E') minus outflows from the country from the rest of the world (M')

 What is Government savings?

Government savings = Net Government income (T) – Government expenditure (or G)

Net Government income is T or Taxes – Transfers from government

So Government savings (S_{gov}) = $T - G$

 From the above, Private Savings (S_{pvt}) + Government savings (S_{gov}) = $I + CA$

 Foreign Savings?

Remember Balance of Payments

Overall Balance of Payments = Current Account Balance + Capital Account Balance

(capital account refers to the net inflow of foreign direct investment, foreign institutional investment, and so on)

If there is an overall balance of payments surplus (the sum of current account and capital accounts), it will be held as reserves by the Central Bank.

India has a deficit in its current account and a surplus in capital account (and overall surplus in its balance of payments, which is reflected in India's large foreign exchange reserves).

China has a surplus in its current and capital accounts and it holds huge forex reserves.

China's reserves (which is held as Chinese deposits in the US government securities) is, in a way, a transfer of Chinese savings to the United States

The United States, which has a large current account deficit, finances its excess consumption through foreign (Chinese/Asian) savings.

Therefore, a deficit in current account (-CA) indicates injection of foreign savings.

We can see from above that

Private Savings (S_{pat}) + Government savings (S_{gov}) - CA = I

Or private investment (I) is financed through Private savings + government savings + foreign savings

$$(I - S_{pvt}) + (G - T) + (E' - M') = 0$$

Why I, G and E' are demand injections?

Why S_{pvt} , T, and M' are demand leakages?

The demand injections and demand leakages are on account of three institutions: the private sector, the government, and the external or the foreign sector.

Investment, government expenditures and exports are three autonomous or independent sources of demand in the economy. They can independently (or on their own) increase expenditures, leading firms to employ more workers to meet the higher production targets, which results in higher incomes and even more jobs.

On the other hand, private savings, taxes, and imports all lead to demand leakages. Savings may be a private virtue (preparing oneself for a “rainy day”). But saving is also delayed consumption, which is delayed expenditure or withheld demand. Savings may be useful as a source of funds for tomorrow’s economy (or the economy five years later). At the same time, today’s savings are a loss of today’s sales for the shopkeeper and the economy.

Therefore, the ‘paradox of thrift’ according to John Maynard Keynes. Saving is a private virtue and good for the individual and the family. But if everybody starts saving a large share of their incomes, especially when the economy is going through a depression, such saving harms the economy. What is good for the individual is not always good for the larger economy and society. That is the paradox.

Saving is critical for generating funds for capital formation and future growth. In the 1950s, one of the main objectives of Indian planning was to raise the country’s saving rate to 20%. However, a very high savings rate can slow economic growth (as happened in the case of Japan and China).

The same is true with government taxes. The taxes collected this year will help build schools and roads in the future (or so one hopes in a democracy), but the amount paid as taxes this year could have been used instead for greater consumption expenditures (to buy more clothes or for more eating out), which could have benefited (the shop owners, the eateries) economic growth this year.

When a country imports goods and services, it loses a part of the domestic demand for foreign industries and businesses. Alternatively, when, say, India exports drugs, Indian firms are gaining a share of the foreign demand.

Some of the major references used for preparing the Notes:

Taylor, Lance (2009), *Reconstructing Macroeconomics*, Harvard University Press.

Macroeconomics by Errol D'Souza, chapters 1 and 2, Pearson Education

Macroeconomics: The Dynamics of Commodity Production by Amit Bhaduri, Macmillan

Some Numbers to illustrate: (One does not have to memorize the numbers! But by going through these tables and figures, one gets a better understanding of the concepts discussed above)

India's macroeconomic aggregates in 2019-20 (at constant prices, with 2011-12 as base year)

	in trillion
1. Gross value added (GVA) at basic prices	132.7
2. Net taxes on products	13.0
3. Gross Domestic Product (1+2)	145.7
4. Consumption of Fixed Capital (or depreciation)	17.5
5. Net Domestic Product (3-4)	128.2
6. Net primary (or factor) incomes received from	-1.4
7. Gross National Income (3 - 6)	144.3
8. Net National Income (7 - 4)	126.8
9. Gross Capital Formation	53.0

Source: India's National Income Statistics available here: <https://mospi.gov.in/web/mospi/reports-publications/-/reports/view/templateFive/901?q=RPCAT>

Sector-wise Distribution of Gross value added, 2018-19, at constant 2011-12 prices: in trillion Rupees and as %,

Sector	in trillion	as %
Agriculture and allied activities	18.9	14.8
Manufacturing	23.3	18.3
Industry	29.5	23.2
Construction	10.3	8.0
Trade, hotels, transport,	25.4	19.9
Finance, real estate, professional	27.2	21.3
Public administration, defence	16.2	12.7
Gross value added (GVA) at basic prices	127.4	100

Source: India's National Income Statistics available here: <https://mospi.gov.in/web/mospi/reports-publications/-/reports/view/templateFive/901?q=RPCAT>

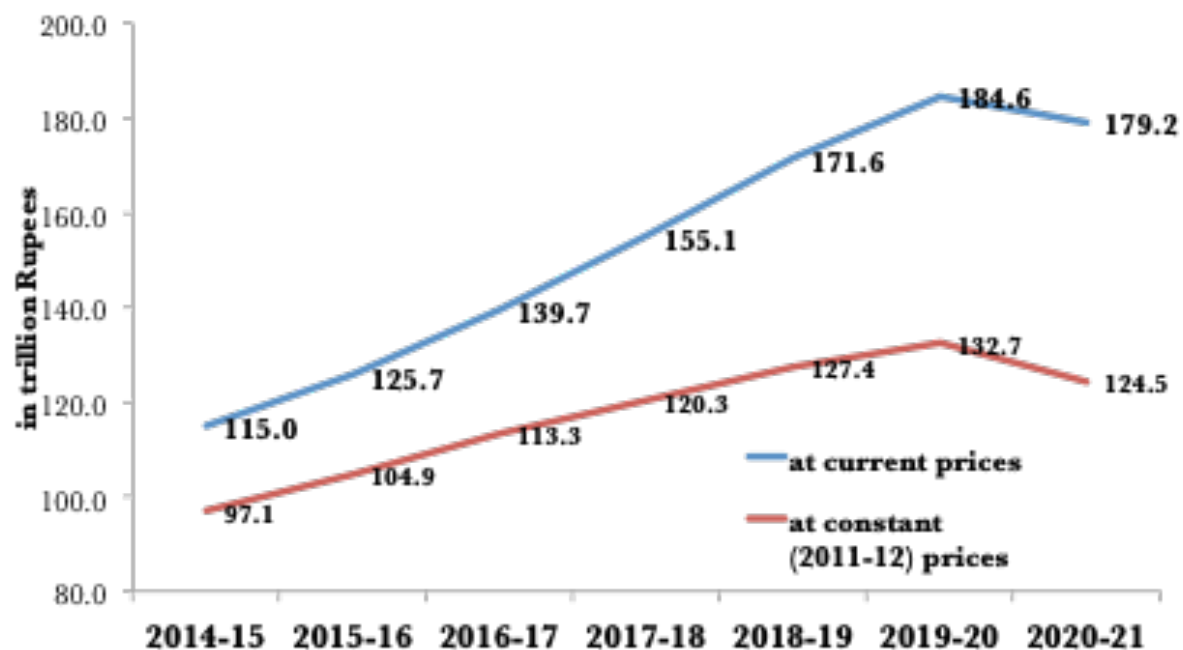
Components of India's GDP at market Prices, 2012-13, at constant 2004-05 prices

	in Rupees billion	As % of GDP at market prices
C =PFCE in the domestic	35695	61
I =Private sector GCF	17811	30
Cg =Government final	6620	11
Ig=Public sector GCF	5087	9
X=exports of goods & services	14498	25
M=imports of goods & services	19896	34
GDP at market prices	58998	100

Source: Reserve Bank of India, Handbook of Statistics on Indian Economy

GVA at basic prices: at current prices and constant (2011-12) prices

Source: National Income Statistics

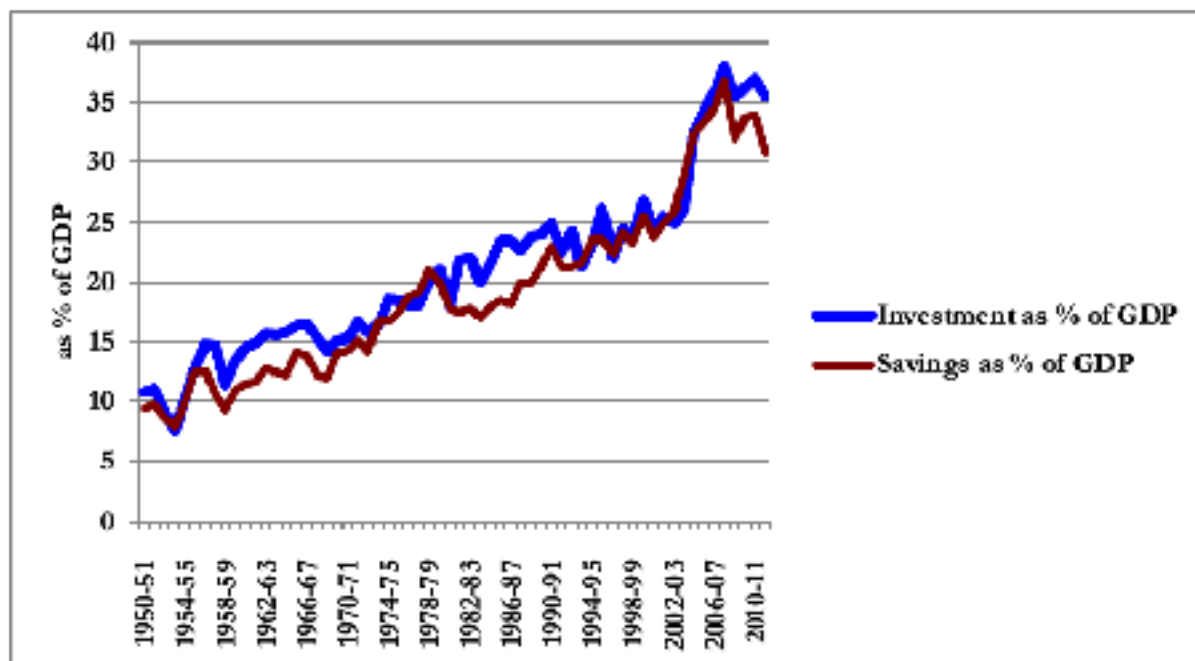


Savings and Investment (gross capital formation) as % of GDP, 2014

	Savings rate	Investment rate
India	31.6	31.6
China	49.5	46.2
East Asia & Pacific (developing)	45.0	41.6
United States	17.6	19.4
United Kingdom	12.4	17.5
Greece	10.8	
Bangladesh	38.9	28.6

Source: World Development Indicators

Savings and Investment as % of GDP, India, 1950-51 to 2010-11



Current Account Deficit and Savings minus Investment in India, as % of GDP, 1950-51 to 2011-12

